Osaid Khan

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Education

Pace University New York City, NY
Master of Science (MS) in Computer Science (GPA - 4.0/4.0) *Jan 2024 - Dec 2025*

Dr. A.P.J. Abdul Kalam Technical University Noida, India
Bachelor of Technology (B.Tech) in Computer Science *Jul 2016 - Sep 2020*

Technical Skills

AI/ML Performance: *workload characterization*, hardware performance analysis, system-level KPIs, memory-bound vs. compute-bound analysis, inference latency, throughput, power efficiency
Architecture: *CPU and SoC architecture*, memory hierarchies, out-of-order execution, vector/SIMD pipelines, DDR/LPDDR bandwidth, on-chip memory, interconnect, data movement latency
AI Acceleration: GPU/NPU hardware, dedicated inference accelerators, DSP pipelines, model quantization, sparsity, operator fusion, memory reuse, cache and scheduling optimization
Compilers & ISA: MLIR, IREE, TVM, TFLite; graph lowering, tiling, scheduling; MIPS/S8200, AVX, NEON, RVV, AMX, SME, RISC-V Vector/Matrix extensions
Languages & Tools: C/C++, Python, CUDA, PyTorch, NVIDIA Nsight, OpenTelemetry, vLLM, TensorRT-LLM, open-source software, Linux, Git, Docker, Kubernetes

Professional Experience

Software Engineer Phoenix, AZ
Sperse *Feb 2026 - Present*

- Executed **workload characterization** for production AI/ML systems by selecting representative model-and-tool paths, defining OpenTelemetry measurement methodology, and projecting latency and throughput **system-level KPIs** for a platform serving 10,000+ users through 40+ agents
- Performed **hardware performance analysis** across compute, memory, and software boundaries with trace-driven experiments, isolating scheduling and data-movement bottlenecks and translating findings into architecture and optimization recommendations
- Built durable parallel execution for AI/ML workloads with Python, FastAPI, Temporal, Redis, and Kubernetes, reducing software overhead through retries, resumability, and live progress while improving high-concurrency stability
- Represented software in cross-functional architecture discussions spanning CPU, SoC, memory, interconnect, compilers, runtimes, and ML frameworks; translated performance tradeoffs into one-page recommendations and detailed technical findings

Software Engineer Noida, India
Qualitest *Feb 2021 - Dec 2023*

- Characterized memory-bound data workloads in a Java/Spring Boot storage service spanning S3, Google Cloud Storage, and MinIO, reasoning quantitatively about latency, utilization efficiency, and system-level memory bandwidth constraints
- Improved memory access and query throughput by optimizing **PostgreSQL** window functions, materialized views, and indexes, then presented performance metrics through React visualizations for business stakeholders
- Orchestrated 100K+ weekly executions with **AWS Step Functions** and Lambda, removing timeout bottlenecks and improving scheduling efficiency, load balance, and failure recovery under production concurrency

Projects

Agentic CUDA Optimization Virtual Lab | *CUDA, C++, Python, LangGraph, NVIDIA Nsight* github.com/ozzyozbourne

- Built a workload-driven optimization loop for handwritten **CUDA** kernels (matrix operations, normalization, activation functions), using evolutionary search and **NVIDIA Nsight** to identify compute throughput, memory access, pipeline, and scheduling bottlenecks

- Implemented operator fusion, vectorization, and precision experiments; benchmarked inference latency, throughput, and GPU memory against **vLLM** and **TensorRT-LLM**, informing HW/SW co-optimization tradeoffs relevant to graph lowering and edge AI acceleration